# Sahaj Bhadja

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## Summary

Software engineer pursuing an MSCS at USC with 2+ years supporting **software deployments**, troubleshooting application issues, and improving QA-backed reliability across data-driven systems, customer-facing workflows, and Windows-adjacent operational environments.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Built **platform services and validation workflows** for 500+ users, reducing request latency from **1.2s to 350ms** while improving issue diagnosis, operational visibility, and responsiveness across production analytics applications.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving monitoring, release confidence, and production support readiness across recurring software workflows.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting software deployments, cross-team issue resolution, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and strengthening QA coverage for frequent releases and product changes.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for user-facing applications.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

**LST Capitals** **India**
*Web Developer Intern* *November 2022 – January 2023*
- Built **Flask booking and payment APIs** for a cross-platform services product, supporting 50K+ booking records while maintaining sub-200ms core flows and reliable state synchronization across user sessions.
- Integrated **payment, coupon, and webhook workflows** with Firebase-backed synchronization, reducing manual handling by **70%** while improving issue traceability and operational reliability for external customer experiences.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable services while improving supportability, implementation ownership, and production debugging workflows.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations through concurrent execution, CI/CD-backed releases, and cost-aware debugging.

## Skills

**Languages**: Python, Java, JavaScript/TypeScript, SQL, C++, Bash
**Applications & Support**: troubleshooting, application configuration, deployment support, documentation, client-facing issue resolution
**Backend & APIs**: Flask, FastAPI, Node.js, REST APIs, ETL pipelines, async workflows
**Databases**: PostgreSQL, MySQL, MongoDB, SQL Server, Firebase
**Cloud & Infrastructure**: AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD, Windows/Linux familiarity
**Testing & QA**: automated testing, validation workflows, regression reduction, release verification

## Projects

**HMIS** Live | *MERN, TypeScript, Twilio, Appwrite*
- Built a **customer-facing scheduling platform** with appointment workflows, notifications, and dashboard experiences, reducing administrative workload by **30%** through secure automation and responsive interfaces.

**FinNexus** Live | *Next.js, TypeScript, MongoDB, Plaid*
- Built a **financial application** supporting secure account linking, real-time transaction visibility, and advanced filtering across 1M+ transactions for 50,000+ users through scalable data-backed workflows.

**Linguistix** Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.